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Transformation of explosive Itô diffusions using gamma and glued asymptotic scale functions

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Abstract

In this thesis, we are interested in studying stochastic differential equations whose solutions explode to infinity in finite expected time.

First, we classify this phenomenon of explosion to infinity for the general form of these differential equations, and then focus on those whose solutions explode in finite expected time, because this type of explosion is in practice most problematic for simulation, for evident numerical reasons.

In establishing conditions for explosion on a certain class of stochastic differential equations, we identify different regimes of explosion and correspondingly construct a transform to scale an explosive process into a non-explosive one that can then be used preferentially for numerical simulation.

Introduction

In this chapter, we introduce the problem along with the important concepts of probability theory and numerical analysis from literature that will be used throughout, beginning with the fundamental concepts of a stochastic process and an Itô diffusion.

Definition 1.1 (Definition 1.1.1 in [10])**.** A real-valued *stochastic process* $X = (X_t)_{t\geq 0}$ is a family of measurable functions from a probability space $(\Omega, \mathscr{F}, \mathbb{P})$ into $(I, \mathscr{B})$, where $I = (\ell, r) \subseteq \mathbb{R}$ and $\mathscr{B}$ is the σ-algebra of Borel sets of I.

Definition 1.2 (Definition 7.2.5 in [10])**.** A real-valued, *time-homogeneous Itô diffusion* is a real-valued stochastic process X defined on a probability space $(\Omega, \mathscr{F}, \mathbb{P})$ and satisfying a stochastic differential equation of the form

$$dX_t = \mu(X_t)dt + \sigma(X_t)dB_t, \tag{$\star$}$$

for $t \geq 0$, where $\mu, \sigma : \mathbb{R} \to \mathbb{R}$ are respectively the *drift* and *diffusion* coefficients of X.

In this thesis, we will study Itô diffusions where μ and σ are C^1 or even just locally Lipschitz with $\sigma(x) \neq 0$, whose state space is an interval $I = (\ell, r) \subseteq \mathbb{R}$ with endpoints $-\infty \leq \ell < r \leq \infty$, and which are *regular*, meaning that starting from any point in the interior of I, any other point in the interior may be reached with positive probability, that is $\mathbb{P}(\tau_z < \infty \mid X_0 = x) > 0$ for all $\ell < x, z < r$, where here τ_z is a hitting time.

Definition 1.3 (Proposition 1.4.6 in [10])**.** Let X be a real-valued stochastic process from $(\Omega, \mathscr{F}, \mathbb{P})$ into $I \subseteq \mathbb{R}$, and let A be a measurable subset of I. The *hitting time* of X to A is the random variable

$$\tau_A^X := \inf\{t \geq 0 \mid X_t \in A\},$$

with by convention $\inf(\emptyset) = \infty$. By abuse of notation, we write τ_a for $\tau_{\{a\}}^X$ when $a \in I$.

Before moving on to formally introducing the problem, and in order to present a simple example of a diffusion for the reader, note that B in $(\star)$ is a Brownian motion, and is an elementary example of a continuous stochastic process, that for example models the motion of pollen particles moved by water molecules, or the motion of a stock price.

Definition 1.4 (Definition 1.4.1 in [10])**.** A *filtration* on the measurable space $(\Omega, \mathscr{F})$ is a family $\mathbb{F} = (\mathscr{F}_t)_{t\geq 0}$ of sub-σ-algebras of $\mathscr{F}$, increasing in the sense that $\mathscr{F}_s \subseteq \mathscr{F}_t$ if $s \leq t$. A measurable space $(\Omega, \mathscr{F})$ with a filtration $\mathbb{F}$ is said to be a *filtered space*.

Definition 1.5 (Theorem 1.1.9 in [10])**.** A one-dimensional *Brownian motion* is a real-valued stochastic process $B = (B_t)_{t\geq 0}$ defined on a filtered probability space $(\Omega, \mathscr{F}, \mathbb{F}, \mathbb{P})$ such that *(1)* $B_0 = 0$ almost surely, *(2)* B has independent increments, meaning for any $0 \leq t_0 < t_1 < \cdots < t_n$, the random variables $B_{t_i} - B_{t_{i-1}}, i = 1, 2, \ldots, n$, are independent, *(3)* B has Gaussian increments, meaning for all $t \geq 0$ and $h > 0$, $B_{t+h} - B_t \sim \mathcal{N}(0, h)$, and *(4)* B is almost surely continuous in t.

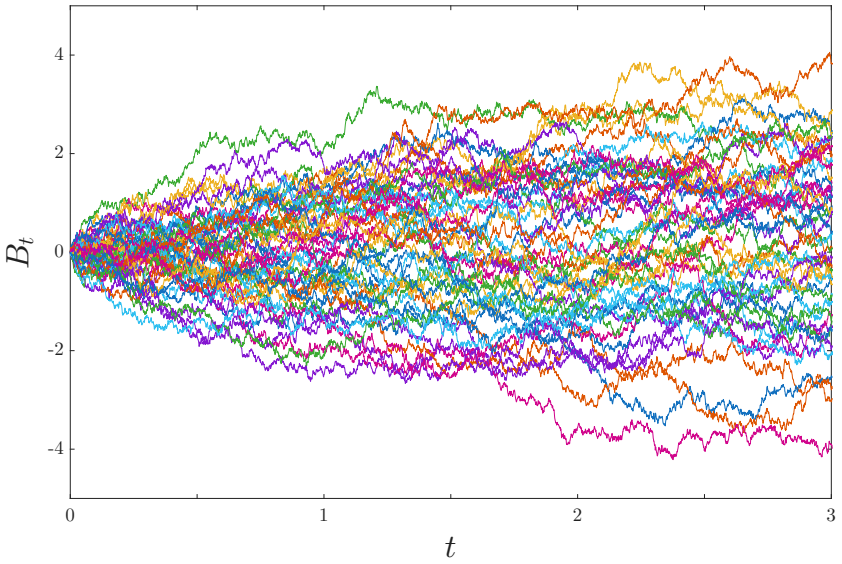


Figure 1.1: Paths of Brownian motion.

As mentioned in the abstract, we are interested in studying Itô diffusions whose solutions explode to infinity in finite expected time. That is, solutions X to $(\star)$ for which certain paths may diverge to infinity as $t \to T < \infty$. This type of explosion is quite problematic for simulation, for evident numerical reasons.

The idea for this thesis is to find a good transform f that could scale an explosive diffusion X into a non-explosive diffusion $M = f(X)$. It is indeed a natural idea to transform a diffusion into another, and for this Itô's lemma is a fundamentally useful identity, serving as a chain rule for stochastic analysis.

Theorem 1.1 (Itô's lemma)**.** *Let X be a time-homogeneous Itô diffusion satisfying $(\star)$, and let $f : I \to \mathbb{R}$ be C^2. Then $f(X)$ is itself an Itô diffusion satisfying*

$$df(X_t) = \left(\mu(X_t)f'(X_t) + \frac{1}{2}\sigma^2(X_t)f''(X_t)\right)dt + \sigma(X_t)f'(X_t)dB_t.$$

As will become clear later after introducing the relevant theory, a driftless diffusion cannot explode to infinity in finite expected time (cf. Remark 1.1), and in the context of our aim of scaling an explosive diffusion X into a non-explosive one, it means that if we can find a function $f \in C^2(\mathbb{R})$ such that

$$\mu(x)f'(x) + \frac{1}{2}\sigma^2(x)f''(x) = 0,$$

then by Itô's lemma the process $M = f(X)$ will be a driftless diffusion satisfying

$$dM_t = \sigma(X_t)f'(X_t)dB_t,$$

and in particular it will not explode in finite time. And in light of the applications to numerical simulation that we aspire to with M, we would additionally like f to be invertible so that not only $M = f(X)$ but also $X = f^{-1}(M)$, and for f to be relatively explicit so that it can actually be useful in practice.

This, then, is the simple idea of this thesis : to find a transform f that is ***(1)*** C^2 on all of $\mathbb{R}$, ***(2)*** invertible, ***(3)*** relatively explicit, and ***(4)*** solves the differential equation for the drift in Itô's lemma. This transform could then be used to for example estimate τ_r^X without having to simulate the explosive process X.

Lemma 1.2. *Let X satisfy $(\star)$ such that X explodes in finite expected time at r, and let f be invertible and C^2 on $\mathbb{R}$. Then $\tau_r^X = \tau_{f(r)}^M$, where $M = f(X)$.*

Proof. By definition $\tau_r^X = \inf\{t \geq 0 \mid X_t = r\}$, and since f is invertible by assumption, then trivially $\tau_{f(r)}^M = \inf\{t \geq 0 \mid f(X_t) = f(r)\} = \tau_r^X$. ■

1.1 Diffusions as transformed Brownian motions

Solving the differential equation for the drift in Itô's lemma is not too complicated, and its solution is for example given in 15.3.6 in [7] as

$$S(x) = \int_{x_0}^{x} s(\xi)d\xi\,, \qquad s(\xi) = \exp\left(-\int_{\xi_0}^{\xi} \frac{2\mu(\eta)}{\sigma^2(\eta)}d\eta\right),$$

for any fixed $\ell < x_0, \xi_0 < r$. This fundamentally important function is appropriately called the *scale function*, since it scales X into a driftless diffusion $M = S(X)$, and since driftless diffusions are local martingales (cf. Definition 4.1.5 in [10]) and therefore in some sense are balanced processes. Since for a fixed diffusion X, its scale function is the unique function to fulfil property *(4)*, S will serve as the basis for our desired transform.

After this change of space, an additional change of time can transform the local martingale $M = S(X)$ into a Brownian motion. For this we need the notion of quadratic variation of a process.

Definition 1.6 (Definition 1.2.3 in [10])**.** Let X be a real-valued stochastic process defined on $(\Omega, \mathscr{F}, \mathbb{P})$. Its *quadratic variation* is the process $\langle X \rangle$ defined by

$$\langle X \rangle_t = \lim_{|\Delta| \to 0} \sum_i (X_{t_{i+1}} - X_{t_i})^2,$$

where Δ is a subdivision $0 = t_0 < t_1 < \cdots < t_n = t$ of $[0, t]$ with mesh $|\Delta|$, and the limit is defined using convergence in probability. If X is an Itô diffusion satisfying $(\star)$, then

$$\langle X \rangle_t = \int_0^t \sigma^2(X_u) du.$$

Theorem 1.3 (Dambis, Dubins–Schwarz)**.** *Let $M = (M_t)_{t \geq 0}$ be a continuous $(\mathscr{F}_t)$-local martingale with $M_0 = 0$ and such that $\langle M \rangle_\infty = \infty$ and let*

$$T_t = \inf\{u : \langle M \rangle_u > t\},$$

then $B = (B_t)_{t \geq 0} = (M_{T_t})_{t \geq 0}$ is a $(\mathscr{F}_{T_t})$-Brownian motion and $M_t = B_{\langle M \rangle_t}$.

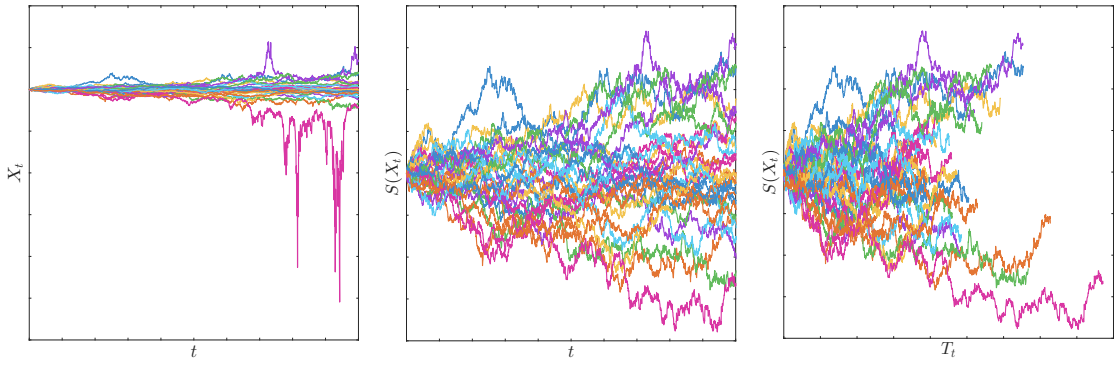


Figure 1.2: Illustration of the Dambis Dubins–Schwartz theorem for $\mu(x) = x(1 + x^2)$ and $\sigma(x) = 3 + x^2$ and $X_0 = 0$.

The random variable T_t at the heart of the above theorem essentially just normalises the clock of M to give a Brownian motion, and it also puts into light another very important function for the study of an explosive diffusion X, namely its *speed density* m. For this we need the notion of the local time of a process and one of its corollaries.

Definition 1.7 (Theorem 6.1.2 in [10])**.** Let X be a continuous semimartingale on $\mathbb{R}$, and let $x \in \mathbb{R}$. The *local time of X in x* is the process $L^x(X)$ defined by

$$L^x(X)_t = |X_t - x| - |X_0 - x| - \int_0^t \operatorname{sgn}(X_u - x) dX_u.$$

Intuitively, $L^x(X)_t$ is an appropriately rescaled and time-parametrised measure of the amount of time X has occupied x up to time t.

Corollary 1.4 (Corollary 6.1.6 in [10])**.** *There is a $\mathbb{P}$-negligible set outside which*

$$\int_0^t f(X_u) d\langle X \rangle_u = \int_{-\infty}^{\infty} f(x) L_t^x(X) dx,$$

for every t and every positive Borel function f.

The formula in Corollary 1.4 is called the occupation time formula, and as mentioned above it puts into light the role of m as a density for the measure of the speed of X in different parts of its state space

$$\begin{aligned} \langle M \rangle_{T_t} \coloneqq t = \int_0^t du &= \int_0^t \frac{d\langle X \rangle_u}{\sigma^2(X_u)} \\ &= \int_{-\infty}^{\infty} \frac{L_t^x(X)}{\sigma^2(x)} dx \\ &= \int_{-\infty}^{\infty} s(x) L_t^x(X)\, m(x) dx\,, \qquad m(x) = \frac{1}{\sigma^2(x) s(x)}. \end{aligned}$$

Equipped with the scale function S and speed density m, we can finally begin to formally study the behaviour of X at a boundary, and hence eventually also formalise the phenomenon of explosion to infinity.

1.2 Behaviour at a boundary

As is done in chapter 15.6 of [7], we here classify the boundary behaviour of an Itô diffusion X defined on $(\ell, r) = (-\infty, \infty)$ satisfying $(\star)$. We concentrate on the right

boundary $r = \infty$, the left boundary begin entirely similar. To ease notation for closed intervals $J = [c, d] \subset (\ell, r)$, we write

$$S[J] = S[c, d] = S(d) - S(c).$$

By its definition as the integral of an exponential, for $\ell < c < x < d < r$,

$$0 < S[c, d] < \infty\,, \qquad \text{and} \qquad S[c, d] = S[c, x] + S[x, d]$$

such that $S[a, b]$ is increasing in b for fixed a and that therefore we may define $S[a, r)$ by

$$S[a, r) = \lim_{b \to r^-} S[a, b] \leq \infty\,, \qquad b < r.$$

In a similar manner, for each path starting at $X_0 = x \in (a, b)$, the hitting time τ_b is an increasing function of b, and therefore we may define the random time τ_{r^-} by

$$\tau_{r^-} = \lim_{b \to r^-} \tau_b \leq \infty\,, \qquad b < r.$$

This leads to a first lemma relating S to the behaviour of X at the boundary r.

Lemma 1.5 (Lemma 15.6.1 in [7])**.** *Let $x_0 \in (\ell, r)$. Then*

- *if $S[x_0, r) < \infty$ then $\mathbb{P}(\tau_a \geq \tau_{r^-} \mid X_0 = x) > 0$ for all $\ell < a < x < r$,*
- *if $S[x_0, r) = \infty$ then $\mathbb{P}(\tau_a > \tau_{r^-} \mid X_0 = x) = 0$ for all $\ell < a < x < r$.*

In light of the above lemma, the boundary r is called *attracting* if $S[x, r) < \infty$, and this criterion applies independently of $x \in (\ell, r)$.

Example 1.1. For a Brownian motion with $\mu = 0$ and $\sigma = 1$ on $(\ell, r) = (-\infty, \infty)$, $r = \infty$ is not attracting since $S[x, b] = b - x$ and $S[x, \infty) = \infty$. On the other hand if instead $\mu = 1 > 0$, then $S[x, b] = e^{-2b} - e^{-2x} \underset{b \to \infty}{\longrightarrow} -e^{-2x} < \infty$, and hence $r = \infty$ is attracting.

The above example shows that an attracting boundary need not be in the state space, and that the probability of reaching an attracting boundary in finite time may be zero. This refinement is achieved by introducing another important function for the study of boundary behaviour

$$\Sigma(r) = \Sigma(r, x) = \int_x^r \left(\int_\xi^r s(\eta) d\eta \right) m(\xi) d\xi = \int_x^r S[\xi, r) dM(\xi),$$

where we have naturally introduced the *speed measure* induced by m as

$$M[J] = M[c,d] = \int_c^d m(x)dx, \qquad J \subset (\ell, r).$$

Note that the double integral for $\Sigma(r,x)$ depends on x, but in our considerations only whether its value is finite or infinite matters, hence why we ignore the dependence on x. And similarly to Lemma 1.5, there is a lemma relating Σ to the behaviour of X at r.

Lemma 1.6 (Lemma 15.6.2 in [7])**.** *Let r be an attracting boundary and $\ell < a < x < r$. Then the following are equivalent*

- $\mathbb{P}(\tau_r < \infty \mid X_0 = x) > 0$,
- $\mathbb{E}\big[\min\{\tau_a, \tau_r\} \mid X_0 = x\big] < \infty$,
- $\Sigma(r) < \infty$.

Heuristically, $\Sigma(r,x)$ measures the time needed for X to reach the boundary r or an alterative interior state a starting from x with $\ell < a < x < r$, by integrating the normalised distance given by S as measured by M, along the path from x to r.

Table 1.1 (Table 15.6.2 in [7])**.** The boundary classification of all possible combinations of assignments of finite or infinite value to $S[x,r)$, $M[x,r)$, $\Sigma(r)$, and $N(r)$.[†]

Criteria				Terminology		
$S[x,r)$	$M[x,r)$	$\Sigma(r)$	$N(r)$	Boundary type	r	
$< \infty^*$	$< \infty^*$	$< \infty$	$< \infty$	regular	attracting	attainable
$< \infty$	$= \infty^*$	$< \infty^*$	$= \infty$	exit		
$< \infty^*$	$= \infty^*$	$= \infty^*$	$= \infty$	natural $\Sigma(r) = \infty^*, N(r) = \infty^*$	nonattracting	unattainable
$= \infty^*$	$< \infty^*$	$= \infty$	$= \infty^*$			
$= \infty^*$	$= \infty^*$	$= \infty$	$= \infty$			
$= \infty^*$	$< \infty$	$= \infty$	$< \infty^*$	entrance		

[†]Minimal sufficient conditions for establishing each combination are indicated by an asterisk.

There is a final quantity $N(r)$ (cf. 15.6.12 in [7]) that conversely to $\Sigma(r)$ measures the time needed for X to reach an interior point $x \in (\ell, r)$ starting at r, that we mention for completeness of the classification.

Remark 1.1. If X is a driftless diffusion, that is if $\mu = 0$, then $S[x,b] = b - x$, and hence $S[x,\infty) = \infty$ and therefore X cannot explode to infinity in finite expected time since necessarily $\Sigma(\infty) = \infty$.

Explosion in finite time of polynomial diffusions

As explained in the introduction, we are here interested in diffusions that explode to infinity in finite expected time, that is, those with $r = \infty$ as an attainable boundary, or in our notation just $\Sigma(\infty) < \infty$, and therefore necessarily also $S[x, \infty) < \infty$.

In the previous chapter, as we classified the phenomenon of explosion of X at r, we found that since its scale function S is the unique function to fulfil property *(4)*, it should a priori be the appropriate transform to obtain a non-explosive diffusion $M = S(X)$. A first problem with S is that, being defined as a nested integral involving an exponential and μ/σ^2, the true scale function S of a diffusion X is in general not explicit at all, and this runs counter to property *(3)* that would want it to be relatively explicit so that it can be useful in numerical applications.

Hence, after establishing conditions for explosion on a certain class of stochastic differential equations, we will study the *asymptotic scale function* S_r of a diffusion X, where we simply replace μ and σ by their asymptotic equivalent in the definition of S. This candidate transform comes as a good compromise between properties *(3)* and *(4)*. And when S_r does not readily also fulfil properties *(1)* and *(2)* of being C^2 and invertible on $\mathbb{R}$, we resort to glueing to obtain a final candidate transform $\tilde{S}$ that combines all properties to the best of our ability.

2.1 Criteria for explosion

Let X be a time-homogeneous Itô diffusion on $(\ell, r) = (-\infty, \infty)$ satisfying $(\star)$ with

$$\mu(x) \underset{+\infty}{\sim} ax^\alpha\,, \qquad \sigma(x) \underset{+\infty}{\sim} bx^\beta, \quad b \neq 0. \tag{$*$}$$

A natural first question is to ask for conditions on a, b, α, β such that $r = \infty$ is an attainable boundary for such asymptotically polynomial diffusions.

Lemma 2.1. *Let X satisfy $(\star)$ with μ, σ as in $(*)$. Then $S[x, \infty) < \infty$ if and only if*

- *$\alpha > 2\beta - 1$ and $a > 0$, or*
- *$\alpha = 2\beta - 1$ and $2a > b^2$.*

Proof. Let $x_0, x_1 \in \mathbb{R}$, and assume first that $\alpha = 2\beta - 1$. Then by definition

$$\begin{aligned} s(x) = \exp\left(-\int_{x_0}^{x} \frac{2\mu(\eta)}{\sigma^2(\eta)} d\eta\right) &\underset{+\infty}{\sim} \exp\left(-\int_{x_0}^{x} \frac{2a}{b^2}\eta^{\alpha-2\beta} d\eta\right) \\ &= \exp\left(-\frac{2a}{b^2}\int_{x_0}^{x} \eta^{-1} d\eta\right) = x_0^{2a/b^2} x^{-2a/b^2}, \end{aligned}$$

and similarly

$$\begin{aligned} S(x) = \int_{x_1}^{x} s(\xi) d\xi &\underset{+\infty}{\sim} x_0^{2a/b^2} \int_{x_1}^{x} \xi^{-2a/b^2} d\xi \\ &= \frac{x_0^{2a/b^2}}{1 - 2a/b^2}(x^{1-2a/b^2} - x_1^{1-2a/b^2}), \end{aligned}$$

from which it is clear that $S[x, \infty) < \infty$ if and only if $1 - 2a/b^2 < 0$, or equivalently if and only if $2a > b^2$. And if $\alpha \neq 2\beta - 1$, then

$$\begin{aligned} s(x) &\underset{+\infty}{\sim} \exp\left(-\int_{x_0}^{x} \frac{2a}{b^2}\eta^{\alpha-2\beta} d\eta\right) \\ &= \exp\left(-\frac{2a}{b^2(\alpha - 2\beta + 1)}(x^{\alpha-2\beta+1} - x_0^{\alpha-2\beta+1})\right) \\ &= A \exp\left(-\frac{2a}{b^2}\frac{x^{\alpha-2\beta+1}}{\alpha - 2\beta + 1}\right). \end{aligned}$$

And therefore

$$S(x) = A\int_{x_1}^{x} \exp\left(-\frac{2a}{b^2}\frac{\xi^{\alpha-2\beta+1}}{\alpha - 2\beta + 1}\right) d\xi,$$

and this converges if and only if $\alpha - 2\beta + 1 > 0$ and $-2a/(b^2(\alpha - 2\beta + 1)) < 0$, or equivalently if and only if $\alpha > 2\beta - 1$ and $a > 0$. ■

Proposition 2.2. *Let X satisfy $(\star)$ with μ, σ as in $(*)$. Then $\Sigma(\infty) < \infty$ if and only if*

- *$\alpha > 2\beta - 1$ (supercritical regime) $\alpha > 1$ and $a > 0$, or*
- *$\alpha = 2\beta - 1$ (critical regime) $\alpha, \beta > 1$ and $2a > b^2$.*

Proof. As mentioned before, $\Sigma(\infty) < \infty$ implies $S[x,\infty) < \infty$, and therefore if suffices to reason over the cases of Lemma 2.1. If $\alpha = 2\beta - 1$, $2a > b^2$, then for $x_0 = 1$

$$m(x) = \frac{1}{\sigma^2(x)s(x)} \underset{+\infty}{\sim} \frac{1}{b^2 x^{2\beta} x^{-2a/b^2}} = \frac{1}{b^2 x^{2\beta - 2a/b^2}}$$

and therefore

$$\begin{aligned}\Sigma(\infty) = \int_x^\infty S[\xi,\infty) m(\xi) d\xi &\underset{+\infty}{\sim} \int_x^\infty \frac{\xi^{1-2a/b^2}}{2a/b^2 - 1} \frac{1}{b^2 \xi^{2\beta - 2a/b^2}} d\xi \\ &= \frac{1}{2a - b^2} \int_x^\infty \xi^{1-2\beta} d\xi\end{aligned}$$

which is finite if and only if $1 - 2\beta < -1$, that is if and only if $\beta > 1$, and therefore $\alpha > 1$ also. And if $\alpha > 2\beta - 1$, $a > 0$, then letting $p = \alpha - 2\beta + 1 > 0$ and $C = 2a/(b^2 p) > 0$, and for $x_0 = 0$,

$$m(x) \underset{+\infty}{\sim} \frac{1}{b^2 x^{2\beta} \exp(-C x^p)}$$

and

$$\begin{aligned}\Sigma(\infty) = \int_x^\infty \left(\int_\xi^\infty s(\eta) d\eta \right) m(\xi) d\xi &\underset{+\infty}{\sim} \int_x^\infty \left(\int_\xi^\infty \exp(-C\eta^p) d\eta \right) m(\xi) d\xi \\ &= \int_x^\infty \frac{1}{pC^{1/p}} \left(\int_{C\xi^p}^\infty t^{1/p - 1} e^{-t} dt \right) m(\xi) d\xi\end{aligned}$$

after the change of variable $t = C\eta^p$. Recognising the definition of the upper-incomplete gamma function Γ and using its asymptotic behaviour as $\xi \to \infty$ (cf. Appendix) gives

$$\begin{aligned}\int_x^\infty \frac{1}{pC^{1/p}} \left(\int_{C\xi^p}^\infty t^{1/p-1} e^{-t} dt \right) m(\xi) d\xi &= \frac{1}{pC^{1/p}} \int_x^\infty \Gamma\big(1/p, C\xi^p\big) m(\xi) d\xi \\ &\underset{+\infty}{\sim} \frac{1}{pC^{1/p}} \int_x^\infty \left((C\xi^p)^{1/p-1} e^{-C\xi^p} \right) \left(b^{-2} \xi^{-2\beta} e^{C\xi^p} \right) d\xi \\ &= \frac{1}{Cpb^2} \int_x^\infty \xi^{1-2\beta-p} d\xi \\ &= \frac{1}{2a} \int_x^\infty \xi^{-\alpha} d\xi\end{aligned}$$

and this is finite if and only if $\alpha > 1$. ■

Remark 2.1. The analysis for a stochastic differential equation is more subtle than for an ordinary one, as explosion of the latter just corresponds to the supercritical regime

where μ essentially dominates σ. But in the stochastic setting it is for example possible to have no explosion even if $\alpha > 1$ and $a > 0$, if β is sufficiently large.

2.2 Asymptotic scaling

The conditions $(*)$ are particularly convenient for finding an explicit form for S, since monomials are easy to integrate and μ/σ^2 will be a monomial when μ and σ are themselves monomials. But for any other non-trivial form of μ and σ, even just polynomials, the true scale function of X is in general not explicit at all.

For this reason, even in our study of diffusions satisfying $(*)$, we will instead study the *asymptotic scale function* S_r of X as our candidate transform, where we will simply replace μ and σ by their asymptotic equivalent, to leverage the convenient properties of monomials and hence get a relatively explicit transform.

Proposition 2.3. *Let X satisfy $(\star)$ with μ, σ as in $(*)$ such that X explodes supercritically in finite expected time at r. Letting $p = \alpha - 2\beta + 1$ and $C = 2a/(b^2 p)$, then the asymptotic scale function of X at r is*

$$S_r(x) = \frac{1}{pC^{1/p}} \gamma\left(\frac{1}{p}, Cx^p\right).$$

where γ is the lower incomplete gamma function (cf. Appendix).

Proof. From Lemma 2.1, and for $x_0 = x_1 = 0$,

$$S_r(x) = \int_0^x \exp(-C\xi^p) d\xi = \frac{1}{pC^{1/p}} \int_0^{Cx^p} t^{1/p-1} e^{-t} dt = \frac{1}{pC^{1/p}} \gamma\left(\frac{1}{p}, Cx^p\right).$$

after the change of variable $t = C\xi^p$ such that $\xi = C^{-1/p} t^{1/p}$. ∎

Proposition 2.4. *Let X satisfy $(\star)$ with μ, σ as in $(*)$ such that X explodes critically in finite expected time at r. Letting $\eta = 2a/b^2$, the asymptotic scale function of X at r is*

$$S_r(x) = \frac{1 - x^{-(\eta-1)}}{\eta - 1}.$$

Proof. Again from Lemma 2.1 and for $x_0 = x_1 = 1$,

$$S_r(x) = \frac{x^{1-2a/b^2} - 1}{1 - 2a/b^2} = \frac{1 - x^{-(\eta-1)}}{\eta - 1}.$$ ∎

2.3 Global scaling

A one-dimensional diffusion can explode at one or both of ℓ and r, and can even explode at ℓ and r under different parameters (p, C, and η) or different regimes, if its coefficients have different asymptotics at its boundaries. For example a diffusion with

$$\mu(x) = \frac{12}{5}x\left(1 + x^2 + \frac{x^4}{1+e^{-x}}\right), \qquad \sigma(x) = 2(1+x^2),$$

explodes supercritically at $\ell = -\infty$ with $p = 2, C = 3/5$ and critically at $r = \infty$ with $\eta = 6/5$. This therefore raises the matter of how to reconcile the asymptotic scaling functions S_ℓ and S_r of a given diffusion in general.

One evident problem is that the asymptotic scale function from Proposition 2.4 for a critically explosive diffusion cannot, as it is, fulfil property *(**1**)* of being C^2 on $\mathbb{R}$, since it has an essential discontinuity at $x = 0$. But because here $(\ell, r) = (-\infty, \infty)$, explosion is spatially asymptotic and hence we can freely resort to glueing our asymptotic scale functions into a global one, by glueing S_ℓ and S_r smoothly and monotonically using patches on a bounded interval containing the origin, thereby also fulfilling properties *(**1**)* and *(**2**)* and obtaining a final candidate transform $\tilde{S}$ that combines all properties.

As we will see, glueing will only be necessary in the critical regime given the asymptotics of γ the lower incomplete gamma function.

Lemma 2.5. *Let X satisfy $(\star)$ with μ, σ as in $(*)$ such that X explodes supercritically in finite expected time at r. Define S_r as in Proposition 2.3. Then S_r is C^2 and increasing on $(0, \infty)$, and for $0 \le i \le 2$,*

$$S_r^{(i)}(0) = x^{(i)}(0).$$

Proof. That S_r is C^2 follows from its definition as the integral of $s \in C^\infty(\mathbb{R})$ defined by $s(x) = \exp(-Cx^p)$, and that it is increasing follows from the fact that s is positive. And using the asymptotic behaviour of γ as $x \to 0$ (cf. Appendix) gives

$$S_r(x) \underset{x\to 0}{\sim} \frac{1}{pC^{1/p}} \cdot p(Cx^p)^{1/p} = x,$$

which indeed implies that $S_r^{(i)}(x) = x^{(i)}(0)$ for $0 \le i \le 2$. ∎

As mentioned above, the essential discontinuity at $x = 0$ of S_r from Proposition 2.4 for a critically explosive diffusion means we need to resort to glueing using a patch on an interval around the origin. If we can do so in a way that the resulting function $\tilde{S}_r$ has the

same properties as the supercritical S_r above, then we will from there be able to define a global candidate transform $\tilde{S}$, by combining $\tilde{S}_\ell$ and $\tilde{S}_r$ smoothly and monotonically at the origin. This is exactly the content of the next lemma, corollary, and proposition.

Lemma 2.6. *Let X satisfy $(\star)$ with μ, σ as in $(*)$ such that X explodes critically in finite expected time at r. Define S_r as in Proposition 2.4, let $\delta > 1$, and define*

$$L_\delta(x) = \lambda_0 + \frac{\lambda_1}{1 + \lambda_2 e^{-\lambda_3 x}}, \qquad R_\delta(x) = \rho_0 + \frac{\rho_1}{1 + \rho_2 e^{-\rho_3 (x-\delta)}},$$

for $\lambda_0 = -S_r(\delta)/2$, $\lambda_2 = 1$, $\lambda_3 = 4/S_r(\delta)$, and $\lambda_1 = S_r(\delta)$ and for

$$\rho_0 = \frac{S_r(\delta)}{2}, \ \rho_2 = \frac{2S_r'(\delta)^2}{2S_r'(\delta)^2 - S_r(\delta)S_r''(\delta)}, \ \rho_3 = \frac{2S_r'(\delta)(1+\rho_2)}{S_r(\delta)\rho_2}, \ \rho_1 = \frac{S_r(\delta)(1+\rho_2)}{2}.$$

Define the fifth-order "smootherstep" function $w(x) = 10\,(x/\delta)^3 - 15\,(x/\delta)^4 + 6\,(x/\delta)^5$, and the glueing function $g_\delta(x) = (1 - w(x))L_\delta(x) + w(x)R_\delta(x)$. Then

(i) *$g_\delta \in C^2(\mathbb{R})$,*

(ii) *$g_\delta^{(i)}(0) = x^{(i)}(0)$ and $g_\delta^{(i)}(\delta) = S_r^{(i)}(\delta)$ for $0 \le i \le 2$,*

(iii) *g_δ is increasing on $(0, \delta)$.*

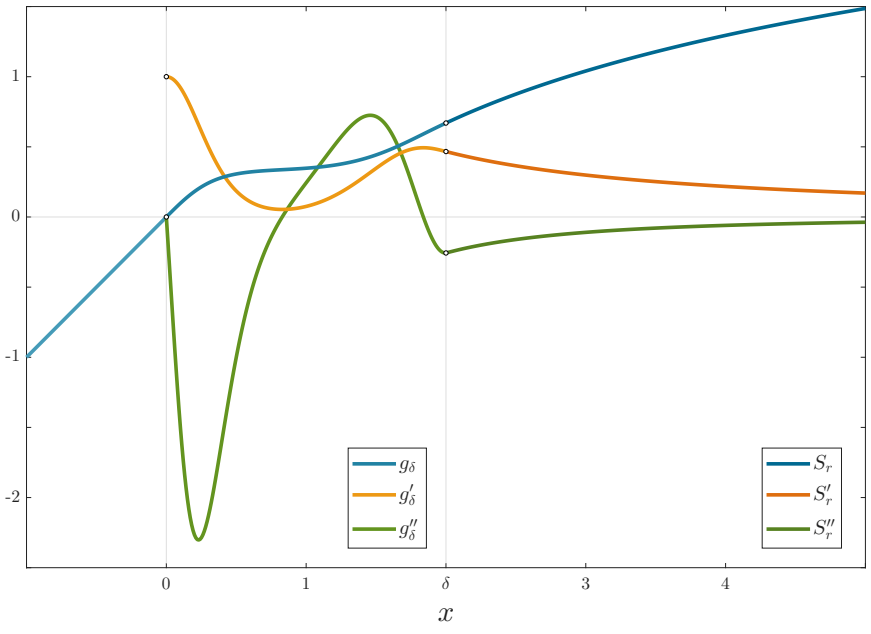


Figure 2.1: The functions $x \mapsto x$, g_δ, S_r and their derivatives for $\eta = 11/10$ and $\delta = 2$.

Proof. ***(i)*** To prove that g_δ is C^2 on $\mathbb{R}$ reduces to proving that the polynomial w and the logistic functions L_δ and R_δ are. This is clearly the case for w, and for L_δ and R_δ this further reduces to proving that $\lambda_2, \rho_2 \geq 0$. In the critical regime

$$S_r(x) = \frac{1 - x^{-(\eta-1)}}{\eta - 1}$$

for $\eta > 1$, and hence

$$\rho_2 = \frac{2\delta^{-2\eta}}{2\delta^{-2\eta} + S_r(\delta)\delta^{-\eta-1}\eta} = \frac{2\delta^{-\eta}}{2\delta^{-\eta} + S_r(\delta)\eta/\delta} > 0,$$

as is the case for $\lambda_2 = 1 > 0$.

(ii) Next, note that

$$g'_\delta = \big[(1-w)L'_\delta + wR'_\delta\big] + w'\big[R_\delta - L_\delta\big],$$
$$g''_\delta = \big[(1-w)L''_\delta + wR''_\delta\big] + 2w'\big[R'_\delta - L'_\delta\big] + w''\big[R_\delta - L_\delta\big].$$

Therefore, since $w(\delta) = 1$ and $w(0)$, $w'(0)$, $w'(\delta)$, $w''(0)$, $w''(\delta)$ are all 0 by construction, this gives exactly

$$g_\delta^{(i)}(0) = L^{(i)}(0) \quad \text{and} \quad g_\delta^{(i)}(\delta) = R^{(i)}(\delta), \qquad\qquad 0 \leq i \leq 2.$$

Hence we just need to prove the second-order equivalence of L_δ to $x \mapsto x$ at 0 and that of R_δ to S_r at δ. Note that

$$R'_\delta(x) = \frac{\rho_1\rho_2\rho_3 e^{-\rho_3(x-\delta)}}{(1+\rho_2 e^{-\rho_2(x-\delta)})^2}, \qquad R''_\delta(x) = \frac{\rho_1\rho_2\rho_3^2 e^{-\rho_3(x-\delta)}(\rho_2 e^{-\rho_3(x-\delta)} - 1)}{(1+\rho_2 e^{-\rho_2(x-\delta)})^3}.$$

Therefore we need

$$S_r(\delta) = \rho_0 + \frac{\rho_1}{1+\rho_2},$$
$$S'_r(\delta) = \frac{\rho_1\rho_2\rho_3}{(1+\rho_2)^2} = (S_r(\delta) - \rho_0)\frac{\rho_2\rho_3}{1+\rho_2},$$
$$S''_r(\delta) = \frac{\rho_1\rho_2\rho_3^2(\rho_2 - 1)}{(1+\rho_2)^3} = S'_r(\delta)\frac{\rho_3(\rho_2-1)}{1+\rho_2} = S'_r(\delta)\frac{S'_r(\delta)(\rho_2-1)}{(S_r(\delta)-\rho_0)\rho_2},$$

and the chosen values for ρ_1, ρ_2, ρ_3 are precisely the ones that solve the above system

from bottom to top after choosing $\rho_0 = S_r(\delta)/2$ as the negative asymptote for R_δ. And for L_δ we need

$$0 = \lambda_0 + \frac{\lambda_1}{1+\lambda_2}, \quad 1 = -\frac{\lambda_0\lambda_2\lambda_3}{1+\lambda_2}, \quad 0 = -\frac{\lambda_2 - 1}{\lambda_0\lambda_2},$$

and again the chosen $\lambda_1, \lambda_2, \lambda_3$ solve the above system after choosing $\lambda_0 = -S_r(\delta)/2$ so that the positive asymptote for L_δ is $\lambda_0 + \lambda_1 = S_r(\delta)/2$.

(iii) Finally, note that *(1)* L_δ and R_δ are increasing if and only if $\lambda_1\lambda_3, \rho_1\rho_3 > 0$, and this is the case since $\rho_2 > 0$ and therefore

$$\rho_1\rho_3 = S_r'(\delta)\frac{(1+\rho_2)^2}{\rho_2} = \delta^{-\eta}\frac{(1+\rho_2)^2}{\rho_2} > 0$$

and $\lambda_1\lambda_3 = 4 > 0$, *(2)* $0 < w < 1$ and $w' > 0$ on $(0, \delta)$, and *(3)* by the deliberate construction of their asymptotes, $L_\delta < S_r(\delta)/2 < R_\delta$. From this it follows that all three terms of g_δ' are positive on $(0, \delta)$, meaning g_δ is increasing. ■

Corollary 2.7. *Let X satisfy $(\star)$ with μ, σ as in $(*)$ such that X explodes critically in finite expected time at r. Define S_r as in Proposition 2.4, let $\delta > 1$, and define*

$$\tilde{S}_r(x) = \mathbf{1}_{[0,\delta)}(x)g_\delta(x) + \mathbf{1}_{[\delta,\infty)}(x)S_r(x).$$

Then $\tilde{S}_r$ is C^2 and increasing on $(0, \infty)$, and for $0 \le i \le 2$,

$$\tilde{S}_r^{(i)}(0) = x^{(i)}(0).$$

Proof. This is a direct consequence of Lemma 2.6 the definition of S_r. ■

For notational convenience, we let $\tilde{S}_r(x) = S_r(x)$ if X explodes supercritically at r, and unless specified otherwise we by convention also let $\tilde{S}_r(x) = x$ if X does not explode in finite expected time at r. And it should be clear that the function $\tilde{S}_\ell$ of a process X can be constructed as the function $\tilde{S}_{-\ell}$ of the process $-X$.

Proposition 2.8. *Let X satisfy $(\star)$ with μ, σ as in $(*)$ such that X explodes in finite expected time at one or both of ℓ and r, and define*

$$\tilde{S}(x) = \mathbf{1}_{(-\infty,0)}(x)\tilde{S}_\ell(x) + \mathbf{1}_{[0,\infty)}(x)\tilde{S}_r(x).$$

Then $\tilde{S}$ is increasing and C^2 on all of $\mathbb{R}$.

Proof. That $\tilde{S}$ is increasing and C^2 on $\mathbb{R} \setminus \{0\}$ is given by Lemma 2.5 and Corollary 2.7. And since in all regimes (nonexplosive, critical, supercritical) we have $\tilde{S}_\ell^{(i)}(0) = x^{(i)}(0)$ and $\tilde{S}_r^{(i)}(0) = x^{(i)}(0)$ for $0 \leq i \leq 2$, then $\tilde{S}$ is also C^2 at 0. ■

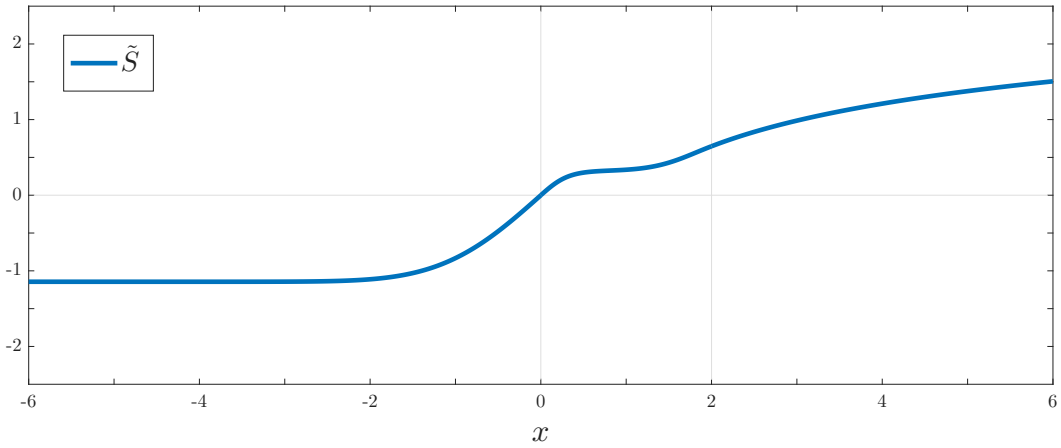


Figure 2.2: For $\mu(x) = 12x(1+x^2+x^4/(1+e^{-x}))/5$ and $\sigma(x) = 2(1+x^2)$, X explodes supercritically at $\ell = -\infty$ with $p = 2, C = 3/5$ and critically at $r = \infty$ with $\eta = 6/5$.

Applications

In this chapter, we go through an example for each regime of explosion and demonstrate how $\tilde{S}$ can be applied to transform diffusions for more stable simulation. But first, we recall the essential method for numerically simulating an Itô diffusion.

Consider as before a diffusion $X = (X_t)_{0 \leq t \leq T}$ satisfying $(\star)$ for $0 \leq t \leq T$ and with initial value $X_0 = x_0$. One of the simplest approximations of this process is the *Euler-Maruyama approximation* (cf. Definition 9.1.3 in [8]). For a given time discretisation $0 = t_0 < t_1 < \cdots < t_n < \cdots < t_N = T$ of the continuous time interval $[0, T]$, an Euler-Maruyama approximation is a continuous time process $Y = (Y_t)_{0 \leq t \leq T}$ satisfying

$$Y_{t_{n+1}} = Y_{t_n} + \mu(Y_{t_n})(t_{n+1} - t_n) + \sigma(Y_{t_n})(B_{t_{n+1}} - B_{t_n}),$$

for $n = 0, 1, \ldots, N-1$ and with initial value $Y_0 = x_0$. In practice it is easiest to partition

the time interval into equidistant time steps $t_n = nh$ for a time step $h = T/N$ for some integer N large enough so that h is very small. The formula arises naturally by writing

$$X_{t_{n+1}} = X_{t_n} + \int_{t_n}^{t_{n+1}} \mu(X_s)ds + \int_{t_n}^{t_{n+1}} \sigma(X_s)dB_s$$

and approximating $\mu(X_s), \sigma(X_s)$ as $\mu(X_{t_n}), \sigma(X_{t_n})$ on the interval $[t_n, t_{n+1}]$. The recursive structure of the approximation, which evaluates values of X at the discretization only, is the reason for its effective implementation on a computer. The conventional way to then determine the value of Y at any time $s \in [0, T]$ is the linear interpolation

$$Y_s = Y_{t_{n_s}} + \frac{s - t_{n_s}}{t_{n_s+1} - t_{n_s}} \left(Y_{t_{n_s+1}} - Y_{t_{n_s}} \right),$$

where n_s is the integer defined by $n_s = \max\{n = 0, 1, \ldots, N : t_n \leq s\}$. This has the advantage that the resulting Y is continuous in t.

Importantly, under a linear growth condition and a Lipschitz condition for μ and σ, an Euler-Maruyama approximation Y converges strongly and weakly to X.

Theorem 3.1 (Theorem 10.2.2 in [8])**.** *Let $X = (X_t)_{0 \leq t \leq T}$ satisfy $(\star)$ on $[0, T]$ and assume that for all $x, y \in \mathbb{R}$, $|\mu(x) - \mu(y)| + |\sigma(x) - \sigma(y)| \leq K_1|x - y|$ for some $K_1 \in \mathbb{R}$ and similarly $|\mu(x)| + |\sigma(x)| \leq K_2(1 + |x|)$ for some $K_2 \in \mathbb{R}$. Then*

- $\mathbb{E}\left[|X_T - Y_T|\right] \leq K_3 h^{1/2} \longrightarrow 0$ *as* $h \to 0$ *for some constant* K_3,
- $\left| \mathbb{E}\left[f(X_T)\right] - \mathbb{E}\left[f(Y_T)\right] \right| \leq K_4 h \longrightarrow 0$ *as* $h \to 0$ *for all continuous* f *and some* K_4,

where h is the time step used in defining Y.

The above theorem and its conditions put into light the intended applications of our transform $\tilde{S}$ defined in Proposition 2.8. If it can indeed scale an explosive diffusion X into another diffusion $M = \tilde{S}(X)$ that fits the conditions of Theorem 3.1, then we will be able to effectively simulate and approximate M, and hence also $X = \tilde{S}^{-1}(M)$. We now demonstrate the application of $\tilde{S}$ in practice in an example for each explosive regime.

3.1 A critically explosive diffusion

Let $c \in \mathbb{R}$ and consider the diffusion X satisfying

$$dX_t = c^2 X_t(X_t^2 + 1)dt + c(X_t^2 + 1)dB_t.$$

This can be solved explicitly as $X_t = \tan(cB_t + \arctan X_0)$, which explodes critically in finite expected time at both $\ell = -\infty$ and $r = \infty$ with $\eta = 2c^2/c^2 = 2$. Its transform glued at $-\delta$ and δ as defined in Proposition 2.8 and by which we define $M = \tilde{S}(X)$ is

$$\tilde{S}(x) = \operatorname{sgn}(x) g_\delta\big(|x|\big)\mathbf{1}_{(-\delta,\delta)}(x) + \operatorname{sgn}(x)\big(1 - |x|^{-1}\big)\mathbf{1}_{(-\infty,-\delta]\cup[\delta,\infty)}(x)$$

such that the state space $(\ell, r) = (-\infty, \infty)$ is mapped to $(-1, 1)$ and that the explosion of X to $r = \infty$ corresponds to M hitting 1, and that $X = \tilde{S}^{-1}(M)$ with

$$\tilde{S}^{-1}(m) = \operatorname{sgn}(m) g_\delta^{-1}\big(|m|\big)\mathbf{1}_{(\frac{1}{\delta}-1,1-\frac{1}{\delta})}(m) + \operatorname{sgn}(m)\big(1 - |m|\big)^{-1}\mathbf{1}_{(-1,\frac{1}{\delta}-1]\cup[1-\frac{1}{\delta},1)}(m).$$

And by Itô's lemma, for $M_t \in (1/\delta - 1, 1 - 1/\delta) \subseteq (-1, 1)$,

$$\begin{aligned} dM_t = &\Big[c^2 \operatorname{sgn}(M_t)\big(g_\delta^{-1}(|M_t|)^2 + 1\big)\big[g_\delta^{-1}(|M_t|) g_\delta'(g_\delta^{-1}(|M_t|)) \\ &\qquad + \frac{1}{2}\big(g_\delta^{-1}(|M_t|)^2 + 1\big) g_\delta''(g_\delta^{-1}(|M_t|))\big]\Big] dt \\ &+ \Big[c\big(g_\delta^{-1}(|M_t|)^2 + 1\big) g_\delta'(g_\delta^{-1}(|M_t|))\Big] dB_t, \end{aligned}$$

and for $M_t \in (-1, 1/\delta - 1] \cup [1 - 1/\delta, 1) \subseteq (-1, 1) \setminus \{0\}$,

$$\begin{aligned} dM_t &= \Big[c^2 X_t (X_t^2 + 1)|X_t|^{-2} - c^2 (X_t^2 + 1)^2 |X_t|^{-3} \operatorname{sgn}(X_t)\Big] dt + \Big[c(X_t^2 + 1)|X_t|^{-2}\Big] dB_t \\ &= \Big[- c^2 \left(1 + \frac{1}{X_t^2}\right) \frac{1}{X_t}\Big] dt + \Big[c \left(1 + \frac{1}{X_t^2}\right)\Big] dB_t \\ &= \Big[c^2 (1 + (1 - |M_t|)^2) \operatorname{sgn}(-M_t)(1 - |M_t|)\Big] dt + \Big[c(1 + (1 - |M_t|)^2)\Big] dB_t. \end{aligned}$$

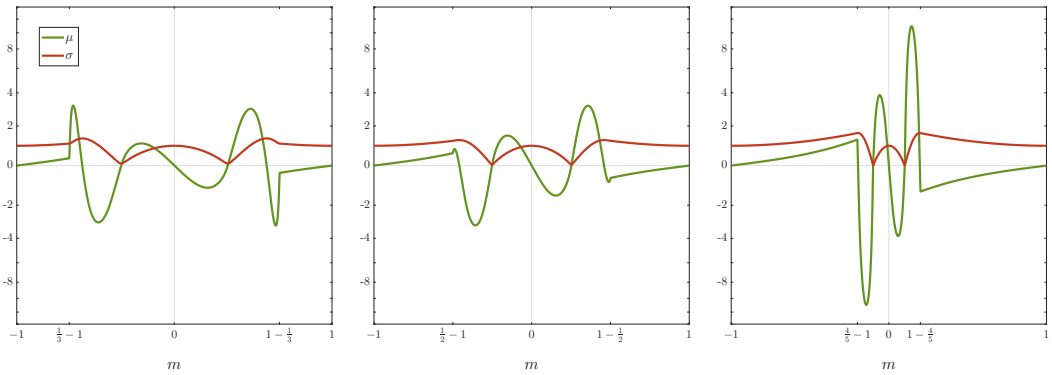


Figure 3.1: The drift and diffusion coefficients μ and σ of M for $c = 1$ and $\delta = 3$, 2, and 5/4.

Here, the choice of $\delta > 1$ considerably affects the geometry of the drift μ and diffusion σ of M. As $\delta \to 1$, the interval of glueing for $\tilde{S}^{-1}$ becomes small, but μ and σ become steep because g_δ becomes flat (and this might be problematic for inverting g_δ numerically).

But with an appropriate choice of delta, any $\delta \geq 2$ for example, we have indeed transformed the explosive process X into a non-explosive process M that we can use preferentially for numerical simulation.

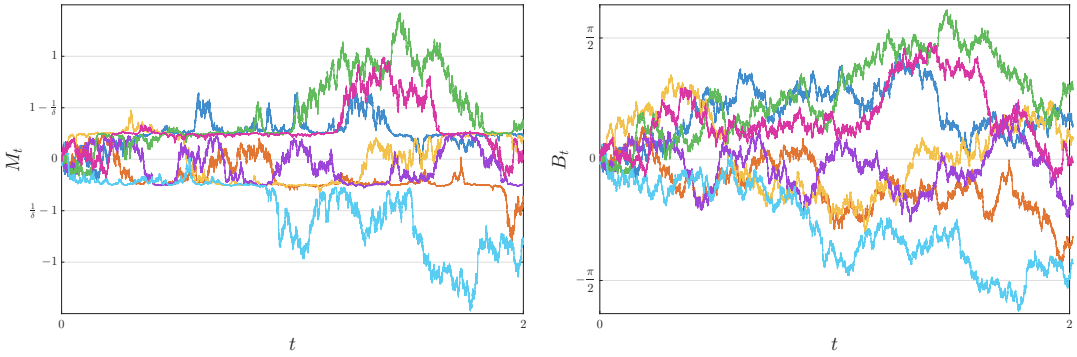


Figure 3.2: Paths of M for $c = 1$ and $\delta = 2$, and of their driving Brownian motion B. B hitting $\pi/2$ corresponds to $X = \tan(B)$ hitting ∞ and M hitting 1.

The transformed process can then be used to more easily estimate the hitting time of X to $r = \infty$, as by Lemma 1.2 this just corresponds to the hitting time of M to $\tilde{S}(\infty) = 1$.

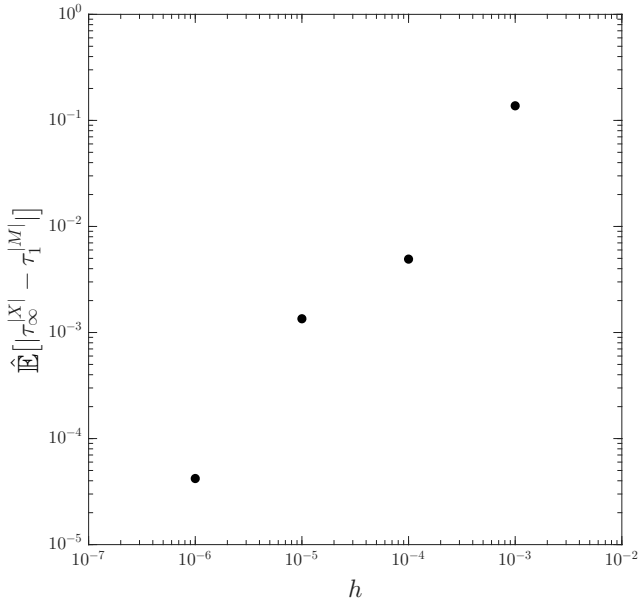


Figure 3.3: Estimation of $\tau_\infty^{|X|}$ using M, as a function of the time step h used in Euler-Maruyama. The empirical expectation $\hat{\mathbb{E}}$ is computed over 100 paths.

3.2 A supercritically explosive diffusion

Consider the diffusion X satisfying

$$dX_t = (X_t^4 + 1)dt + (X_t^2 + 1)dB_t.$$

X explodes supercritically at $r = \infty$ with $p = 4-2\cdot 2+1 = 1$ and $C = (2\cdot 1)/(1^2\cdot 1) = 2$. Its transform by which we define the diffusion $M = \tilde{S}(X)$ is

$$\tilde{S}(x) = \frac{1}{2}\gamma(1, 2x) = \frac{1}{2}\int_0^{2x} e^{-t}dt = \frac{1}{2}\left(1 - e^{-2x}\right),$$

such that the state space $(\ell, r) = (-\infty, \infty)$ is mapped to $(-\infty, 1/2)$ and that the explosion of X to $r = \infty$ corresponds to M hitting $1/2$, and that $X = \tilde{S}^{-1}(M)$ with

$$\tilde{S}^{-1}(m) = -\frac{1}{2}\ln(1 - 2m).$$

And by Itô's lemma

$$\begin{aligned} dM_t &= \left[(X_t^4 + 1)e^{-2X_t} + \frac{1}{2}(X_t^2 + 1)^2(-2e^{-2X_t})\right]dt + \left[(X_t^2 + 1)e^{-2X_t}\right]dB_t \\ &= \left[-2X_t^2 e^{-2X_t}\right]dt + \left[(X_t^2 + 1)e^{-2X_t}\right]dB_t \\ &= \left[\left(M_t - \frac{1}{2}\right)\ln^2(1 - 2M_t)\right]dt + \left[(1 - 2M_t)\left(\frac{1}{4}\ln^2(1 - 2M_t) + 1\right)\right]dB_t. \end{aligned}$$

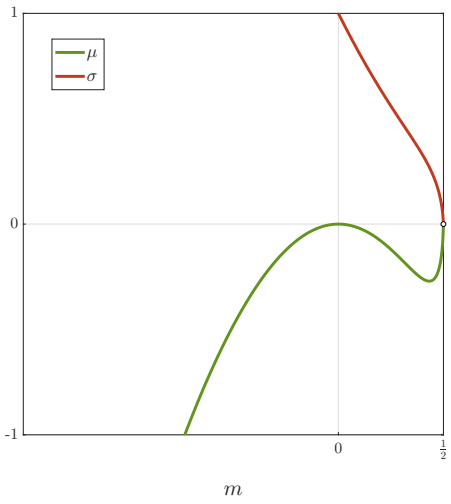


Figure 3.4: The drift and diffusion coefficients μ and σ of M.

Again the explosive process X has been transformed into a non-explosive process M that can be used preferentially for simulation, though here at the cost of the Lipschitz continuity of the drift and diffusion coefficients of M.

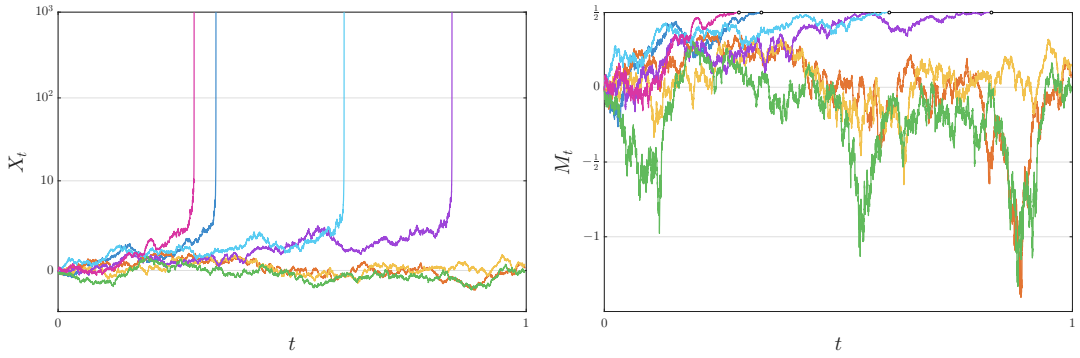


Figure 3.5: Paths of X and of their corresponding M.
M hitting $1/2$ corresponds to X hitting ∞.

Note the contrasting behaviour of X and M near their respective boundary. Compared to the original process X that violently explodes to $r = \infty$, the transformed process M gently gets absorbed at $\tilde{S}(r) = 1/2$, where $\mu(m), \sigma(m) \to 0$ as $m \to 1/2^-$. This suggests the need for a more careful method to estimate τ_r^X using M, as is often needed when dealing with diffusions with non-Lipschitz coefficients.

Conclusion

The methodology and its resulting transform $\tilde{S}$ developed here for scaling polynomial Itô diffusions that explode in finite expected time into non-explosive diffusions combines properties ***(1)***-***(4)*** to the best of our ability, and has the advantage that it should be possible to extend the result of Proposition 2.8 to other classes of explosive diffusions, resorting if necessary to glueing as used in Lemma 2.6. Though this is often hindered by the difficulty in finding relatively explicit forms for asymptotic scale functions.

But for polynomial diffusions, it appears that the applications may be fruitful. For these diffusions, it may be interesting to carry on and find conditions on an explosive

diffusion X satisfying $(\star)$ with μ, σ as in $(*)$ such that the transform $\tilde{S}$ as defined in Proposition 2.8 indeed scales X into a non-explosive diffusion $M = \tilde{S}(X)$, and studying time transformations to complement the scaling of $\tilde{S}$. It may also be interesting to dedicatedly investigate approximations of τ_r^X using $\tau_{\tilde{S}(r)}^M$, and similarly strong approximations of X, possibly in a setting general enough to include non-Lipschitz coefficients such as those from the above supercritical example as is done in [4] and [5] for example.

Appendix

A.1 The gamma function

The *gamma function* can be defined, for $z \in \mathbb{C}$ with positive real part, by the integral

$$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t}\, dt\,, \qquad \mathrm{Re}(z) > 0,$$

and then defined on all of $\mathbb{C}$ as the analytic continuation of this integral function. It is meromorphic on $\mathbb{C}$ with simple poles at zero and every negative integer. It is a very useful special function, and for example $x \mapsto \Gamma(x+1)$ interpolates the points $(n, n!)$ of the factorial sequence for all $n \in \mathbb{N}$. For a complex parameter $z \in \mathbb{C}$ of positive real part and $x \geq 0$, one can then define the *lower incomplete gamma function*

$$\gamma(z, x) = \int_0^x t^{z-1} e^{-t}\, dt,$$

and the *upper incomplete gamma function*

$$\Gamma(z, x) = \int_x^\infty t^{z-1} e^{-t}\, dt.$$

There is a large literature about the gamma function [3] and hence also about its incomplete counterparts. In particular, asymptotically,

$$\gamma(z, x) \underset{x \to 0}{\sim} \frac{x^z}{z} \qquad \text{and} \qquad \Gamma(z, x) \underset{x \to \infty}{\sim} x^{z-1} e^{-z}.$$

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